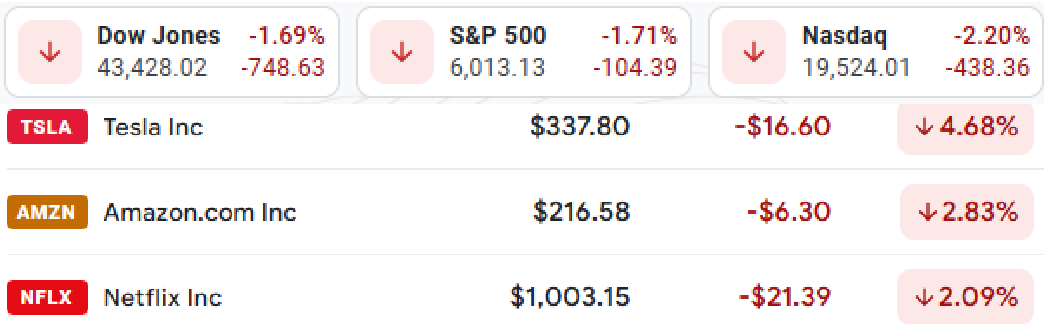
Lab Project: Data Warehouse for Financial Markets Data

You will build a **working data platform** for a fictional company (Acme Ltd) that wants to collect financial market data, keep it trustworthy over time, and use it to generate insights, risk views, and recommendations for customers. The next section will the describe the domain (Financial Markets and Instruments) you will be working with. The Project Description section focuses on defining the project scope and objects.

Introduction to Financial Instruments

Financial markets - where traders buy and sell assets. Assets include stocks, bonds, derivatives, foreign exchange, and commodities. The markets are where businesses go to raise cash to grow. It's where companies reduce risks and investors make money. In this chapter the few popular classes of financial instruments are briefly introduced.

Stocks allow you to own a portion of a public corporation. The owners sell control of the company to stockholders to gain additional funds to grow the company. This is called the initial public offering. After the IPO, the stockholders can resell the shares on the Stock Exchange Markets (e.g. NYSE, BVB).



Dow Jones	43,428.02	-748.63	-1.69%
S&P 500	6,013.13	-104.39	-1.71%
Nasdaq	19,524.01	-438.36	-2.20%

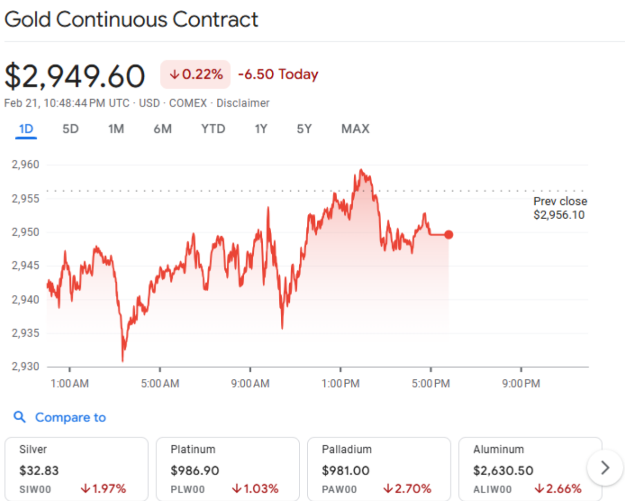
TSLA	Tesla Inc	$337.80	-$16.60	↓4.68%
AMZN	Amazon.com Inc	$216.58	-$6.30	↓2.83%
NFLX	Netflix Inc	$1,003.15	-$21.39	↓2.09%

Bonds are loans made to large organizations. These include corporations, cities, and national governments. An individual bond is a piece of a massive loan. That's because the size of these entities requires them to borrow money from more than one source. Bonds are a type of fixed-income investment.

ISIN	Tip	Data emisiunii/ redeschiderii	Data scadenței	Maturitate reziduală (luni)	Valoare licitație referință (mil.)			Valoare sesiune of. necomp. (mil.)			Monedă	Randament mediu de adjudecare (% p.a.)	Randament maxim de adjudecare (% p.a.)	Rată cupon (% p.a.)
					Prospect	Subscrisă	Adjudecată	Prospect	Subscrisă	Adjudecată				
ROAXXDU0HWO7	CT discont	24 feb.2025	28 ian.2026	11	800,0	1 726,5	1 120,0	-	-	-	lei	6,64	6,65	-
RO52CQA3C829	OS dobândă	24 feb.2025	29 sep.2032	91	600,0	1 609,4	775,0	90,0	130,0	90,0	lei	7,43	7,43	8,25
RO45DLJ4EE76	OS dobândă	19 feb.2025	28 apr.2027	26	500,0	1 035,7	675,7	75,0	200,0	75,0	lei	7,05	7,06	7,40
RO0DU3PR9NF9	OS dobândă	17 feb.2025	24 feb.2038	156	600,0	1 194,1	600,0	90,0	80,0	80,0	lei	7,44	7,48	7,90

A **derivative** is a financial contract that derives its value from an underlying asset. The buyer agrees to purchase the asset on a specific date at a specific price. Derivatives are often used for **commodities**, such as oil, gasoline, or gold. Another asset class is currencies, often the U.S. dollar. There are derivatives based on stocks or bonds. Still others use interest rates, such as the yield on the 10-year Treasury note. Fraud is rampant in the derivatives market (Ponzi scheme on derivatives).

Commodities are raw materials used to manufacture consumer products. They are inputs in the production of other goods and services, rather than finished goods sold to consumers. Investors and traders can buy and sell commodities directly in the spot (cash) market or via derivatives such as futures and options. Hard commodities refer to energy and metals products, while soft commodities are often agricultural goods.



A **cryptocurrency** or colloquially, crypto, is a digital currency designed to work through a computer network that is not reliant on any central authority, such as a government or bank, to uphold or maintain it. Individual coin ownership records are stored in a digital ledger or blockchain, which is a computerized database that uses a consensus mechanism to secure transaction records, control the creation of additional coins, and verify the transfer of coin ownership

#	Name		Price	1h %	24h %	7d %	Market Cap	Volume(24h)	Circulating Supply	Last 7 Days
1	Bitcoin BTC	Buy	$65,693.04	▲0.47%	▼1.32%	▼2.80%	$1,313,636,611,276	$39,704,917,437 604.92K	19.99M BTC	
2	Ethereum ETH	Buy	$1,928.84	▲0.60%	▼1.31%	▼1.08%	$232,796,970,657	$23,040,914,299 11.96M	120.69M ETH	
3	Tether USDT	Buy	$1.00	▼0.00%	▼0.02%	▲0.05%	$183,653,523,824	$81,703,201,917 81.70B	183.63B USDT	
4	BNB BNB	Buy	$613.27	▲0.13%	▼0.56%	▲0.00%	$83,625,155,988	$1,857,082,167 3.02M	136.35M BNB	
5	XRP XRP	Buy	$1.35	▲1.09%	▼1.63%	▼2.50%	$82,829,815,280	$3,344,366,941 2.46B	61.09B XRP	

Financial Instruments Indicators

As one can see from the previous section, there are different indicators/properties/attributes tracked for different types of financial instruments. Some are shared across multiple types of financial instruments, some are specific. This makes from the financial instrument a *heterogeneous* entity of our system model. Below are examples of common stock market properties:

- Current Price / **Market Price** - is the real-time price of a security trading on an exchange, as well as the most recent price of a security listed in a portfolio.
- **Opening Price** - is the price at which a security first trades upon the opening of an exchange on a trading day. The price of the first trade for any listed stock is its daily opening price.
- **Closing Price** - is the final price at which a security is traded on a given trading day. The closing price represents the most up-to-date valuation of a security until trading commences again on the next trading day. Most financial instruments are traded after hours (although with markedly smaller volume and liquidity levels), so the closing price of a security may not match its after-hours price.
- **Adjusted Closing Price** - is a stock's closing price amended to include any distributions and corporate actions before the next day's open.
- **Quoted Price** - is the most recent price at which an investment (or any other type of asset) has traded
- **Ask Size** - is the amount of a security that a market maker is offering to sell at the ask price.
- **Volume** - is the amount of a security that was traded during a given period of time.

Financial Data Vendors

A financial data vendor is an organisation (company) that provides market data to financial firms, traders, and investors. The data distributed is collected from sources such as a stock exchange feeds, brokers and dealer desks or regulatory filings (e.g. an SEC filing).

For example, **Bloomberg** market data provides access to 35 million instruments across all asset classes, aggregated from 330+ exchanges and 5,000+ contributors, as well as Bloomberg composite tickers and market indices. **Nasdaq Data Link**[1] is another provider of market data, having both open source and commercial offerings. Open source data is usually historical and limited to some time frames and for selected instruments. The financial market data is provided in the form of time series, which are described in the next section.

Time Series

A time series is a sequence of numerical data points in successive order. In investing, a time series tracks the movement of the chosen data points, such as a security's price, over a specified period of time with data points recorded at regular intervals. There is no minimum or maximum amount of time that must be included, allowing the data to be gathered in a way that provides the information being sought by the investor or analyst examining the activity.

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[1] https://data.nasdaq.com/institutional-investors

The table below is an example of time series data for General Motors Co. for a period of 5 days. Data provided by Nasdaq.

NOTE: Different data providers include in different indicators in their time series. The time series carry different indicators/properties/attributes for the same type of financial instrument when coming from different providers!

	Open	High	Low	Close	Volume	Ex-Dividend	Split Ratio	Adj. Open	Adj. High	Adj. Low	Adj. Close	Adj. Volume
Date												
2010-11-18	35.00	35.99	33.89	34.19	457044300.0	0.0	1.0	29.988317	30.836558	29.037259	29.294302	457044300.0
2010-11-19	34.15	34.50	33.11	34.26	107842000.0	0.0	1.0	29.260029	29.559912	28.368948	29.354278	107842000.0
2010-11-22	34.20	34.48	33.81	34.08	36650600.0	0.0	1.0	29.302870	29.542776	28.968714	29.200053	36650600.0
2010-11-23	33.95	33.99	33.19	33.25	31170200.0	0.0	1.0	29.088668	29.122940	28.437493	28.488901	31170200.0
2010-11-24	33.73	33.80	33.22	33.48	26138000.0	0.0	1.0	28.900170	28.960146	28.463197	28.685967	26138000.0

Figure 1. General Motors (GM) data from Nasdaq Data Link

Project Scope and Objectives

Through this project our company (Acme Ltd) aims to design and implement a data warehouse that would enable us to extract useful insights from financial data and provide risk assessment and financial recommendations to our customers. The platform must make this data easy to ingest, store, explore, and analyse, while keeping track of where the data came from.

NB: The financial instruments when added to a portfolio they become **financial assets**. A portfolio can be owned by a company or a person. From now on, the terms financial instrument and financial asset may be used interchangeably throughout this project.

Overall Requirements

- Efficiently store and retrieve large amounts of financial data.
- Efficiently feed the data analytics (data mining) tool chain with data.
- Aggregate and integrate data from different financial data vendors (such as Nasdaq Data Link, Bloomerg, etc.)
- DWH will support temporal database paradigm:
    - Existing records in the database cannot be updated, nor deleted,
    - 'Updating' will add instead a new record,
    - 'Deletion' works as adding a special *marker record* marking that starting with the specified valid date the financial instrument is not available anymore.
- Handle heterogenous data, from various internal and external source, the multitude of financial instruments we support will have different sets of attributes.

Common attributes of financial assets (instruments) that needs to be stored and retrieved:

- financial instrument class: stock, bond, crypto currency, interest rate, options, indexes, futures, metals etc.
- symbol: TSL, MSFT, BTC etc.
- description of the financial instrument
- region of origin: US, Europe, China, Africa etc.

- financial data will come in time series; for example, each data point in the series will have an open / close / high price, RON to USD exchange rate, etc.
- other attributes that are unknown now, but may be supported in the future

Functional Requirements (Use Cases)

UC 1. Data Ingest from Financial Data Providers

The system must be able to import data from external providers (e.g., Nasdaq, Bloomberg) through the publicly APIs (usually RESTful API) made available by providers. The source of data (i.e. the provider / provenance) must be recorded such that we're able to trace data provenance in our system.

UC 2. Expose Financial Data for Consumption via RESTful API

Design and implement a RESTful API to allow clients to consume data from our data warehouse. It should allow clients to discover financial instruments, inspect details, and retrieve time series data in a streamlined way. The following are the most frequent queries that need to be efficiently supported by the system:

- [Q1] Return limited info (e.g. identification data, such as assetId) about all financial assets available in the data warehouse
- [Q2] Return all the details of an asset knowing it's identifier
- [Q3] Return limited info (e.g. identification data, such as dataSourceId) about all sources of (financial) data available in the data warehouse (supported data providers)
- [Q4] Return all the details of a financial time-series' data source knowing it's identifier
- [Q5] Return time-series data for specified asset and data source identifiers

UC 3. Enable Data Aggregation, Analytics and Data Mining

Provide clients with means to analyse the data stored in the data warehouse in an effective and flexible approach. The clients need to get insights and forecasts out of the data in the data warehouse. Examples of useful outputs include data trends, comparisons, aggregations (e.g. count/min/max/average), simple forecasts (e.g., next day price), or risk-related signals. Design should make easy to feed an analytics / machine learning tool (e.g., Apache Spark) with data from the platform.

UC 4. Integration with Large Language Models (LLM)

The platform must include an **LLM-powered assistant** that can help a user explore and understand the data in natural language. Integrate the assistant using **MCP**, so the LLM can call your platform's capabilities as tools (for example: "list assets", "fetch time series", "summarize trends", "compare two assets", "explain a change"). The assistant must produce answers grounded in the platform's data, not generic finance text.

Bonus (Optional): Agentic AI, agent-style usage, meaning the assistant can plan multi-step actions (for example: find assets → fetch time series → compute summary → explain the findings). This may use LangFlow support by delivering at least one LangFlow flow that uses the MCP-connected tools (or an equivalent agent workflow that you can demonstrate clearly).

Non-Functional Requirements

The platform must model financial and store assets and linked time series data (prices/indicators over time) for those assets. You may choose any programming language/frameworks, cloud/local setup, and architecture you want. **A NoSQL database as storage system is mandatory**.

The data model/design **must handle heterogeneous data**, meaning different asset classes and providers may have or provide different attributes.

Support a temporal data warehouse approach. Once stored, records must not be overwritten or removed in-place. Changes must be represented by adding new versions, and "deletion" must be represented by a marker that the asset/data is no longer valid from a given moment. The system must be able to return consistent and correct historical data at any given time (present, or past).

Each student **MUST** fill in the **IIAGen (Generative AI Tools) usage statement**, using the template uploaded in the classroom, covering all tasks and activities performed in the scope of this project design/implementation/testing/demo etc.

Deliverables

- A runnable implementation of the platform (local or deployed), with clear run instructions.
- A short project report explaining what you built, what data you used, and how to reproduce your results.
- A short demo video showing ingestion → storage → API exploration → analytics → LLM assistant via MCP end-to-end process. During the demo make sure you use a small set of example queries/prompts that demonstrate the platform's value (both API usage and LLM usage).

Evaluation Criteria (High Level)

- **Completeness:** all mandatory capabilities are present.
- **Correctness:** data, provenance, and temporal history behave as described.
- **Usability:** your API and LLM assistant make the platform easy to explore.
- **Industry-level readiness:** the system looks like something a team could extend, not just a script.